

Radiative Corrections with `aao_rad` and `aao_norad`: An Example of the Calculation as a Function of ϕ

V. Kubarovsky

July 21, 2026

Abstract

This note describes how to use the two versions of the `aao_gen` event generator, `aao_rad` (with internal radiative effects) and `aao_norad` (Born-level, non-radiative), to calculate the radiative correction factor for exclusive single-pion electroproduction. As a working example, the correction is evaluated as a function of the azimuthal angle ϕ of the hadronic plane. The same procedure applies to any kinematic variable (Q^2 , W , t , $\cos\theta^*$, ...).

1 The generators

The repository `aao_gen` contains two Fortran generators:

- `aao_norad` — generates events according to the Born (non-radiative) cross section of the chosen physics model (A0, MAID98, MAID2000);
- `aao_rad` — generates events including internal radiative effects (vertex corrections, vacuum polarization, and real photon emission) following the Mo–Tsai formalism applied to the same underlying model.

Both generators use accept–reject sampling, so the output events are *unweighted*. At the end of the run each generator prints the **integrated cross section** over the generation region,

`Integrated cross section = sig_int sig_sum`

computed by Monte Carlo integration as

$$\sigma_{\text{tot}} = \frac{\sum_k \sigma_k}{N_{\text{tries}}} (4\pi)^2 (2\pi \Delta Q^2 \Delta E'), \quad (1)$$

i.e. the total cross section *within the kinematic limits specified in the input file* (Q^2 range, scattered-electron energy range, W range).

2 From event samples to differential cross sections

Since the events are unweighted, a sample of N events together with the integrated cross section σ_{tot} fully defines the differential cross section in any variable. For a histogram in ϕ with bin width $\Delta\phi$ and n_i events in bin i ,

$$\left. \frac{d\sigma}{d\phi} \right|_i = \sigma_{\text{tot}} \frac{n_i}{N \Delta\phi}. \quad (2)$$

3 The radiative correction factor

The radiative correction factor in bin i is defined as the ratio of the radiated to the Born differential cross sections,

$$RC(\phi_i) = \frac{d\sigma_{\text{rad}}/d\phi}{d\sigma_{\text{norad}}/d\phi} = \frac{\sigma_{\text{rad}}}{\sigma_{\text{norad}}} \cdot \frac{n_i^{\text{rad}}/N_{\text{rad}}}{n_i^{\text{norad}}/N_{\text{norad}}}. \quad (3)$$

For equal statistics, $N_{\text{rad}} = N_{\text{norad}}$ (e.g. 100 000 events each), this reduces to the simple prescription:

1. fill the `aao_norad` histogram $h_{\text{norad}}(\phi)$ with weight 1;
2. fill the `aao_rad` histogram $h_{\text{rad}}(\phi)$ with the constant weight

$$w = \frac{\sigma_{\text{rad}}}{\sigma_{\text{norad}}}, \quad (4)$$

where σ_{rad} and σ_{norad} are the two integrated cross sections printed at the end of the runs;

3. divide bin by bin:

$$RC(\phi) = \frac{h_{\text{rad}}(\phi)}{h_{\text{norad}}(\phi)}. \quad (5)$$

The overall normalization $\sigma_{\text{rad}}/\sigma_{\text{norad}}$ carries the bulk of the correction (vertex correction, vacuum polarization, soft-photon emission), while the bin-by-bin shape ratio adds the kinematic migration caused by hard-photon emission.

4 The internal missing-mass cut in `aao_rad` (`mm_cut`)

The `aao_rad` input file contains the parameter `mm_cut` (the line labeled “a limit on the error in (mm)**2”), which is applied *inside the generator, during event generation*:

```
if (abs(mm2 - mm_exp) .gt. mm_cut) go to 20      ! reject, regenerate
sig_tot = sig_tot + sigr                         ! only surviving events
count
```

Here `mm2` is the *experimental* missing mass squared, computed the way an experimentalist computes it — from the nominal beam energy, the scattered electron, and the detected hadron (the proton for π^0/η , the pion for π^+), *ignoring the radiated photon*. For the π^0 channel it is $MM^2(ep \rightarrow e'pX)$. The reference value `mm_exp` is the nominal missing mass squared of the channel: $m_{\pi^0}^2$ for `epirea=1`, $m_p^2 \simeq m_n^2$ for `epirea=3`, m_η^2 for `epirea=5`. When a hard photon is emitted, `mm2` shifts away from `mm_exp`; this shift *is* the radiative tail.

Since the rejection happens *before* the cross section is accumulated, **the integrated cross section σ_{rad} printed at the end of the run is the cross section within the window $|MM^2 - MM_{\text{exp}}^2| < \text{mm_cut}$** . The parameter therefore plays exactly the role of the inelasticity cut v_{cut} in other radiative-correction codes: it defines how much of the radiative tail the sample — and σ_{rad} — contains, and hence the size of the correction itself.

Practical rules:

1. The radiative correction is only defined *with respect to a specific missing-mass cut*; `mm_cut` must be consistent with the MM^2 cut applied to the data.

2. Recommended: set `mm_cut` *looser* than the analysis cut (the data MM^2 is smeared by detector resolution, so events migrate across the cut boundary in both directions), and apply the exact analysis cut offline on the *reconstructed* MM^2 after GEMC, with the same code as for the data. The normalization stays correct because $d\sigma = \sigma_{\text{printed}} \times n_{\text{pass}}/N_{\text{generated}}$. Do not set it absurdly loose, or most of the generated events fall in the tail that the analysis cut later removes.
3. **Never** set `mm_cut` tighter than the data cut: the part of the radiative tail present in the data would then be missing from the Monte Carlo, and the correction would be underestimated with no way to recover it offline.
4. `aao_norad` has no radiated photon, so at generator level its MM^2 is exact and no such parameter exists there. After detector smearing, apply the same reconstructed- MM^2 cut to both samples.

5 Applying the correction to data

The experimental (radiated) cross section is corrected to the Born level by dividing by the RC factor evaluated in the same bin with the same cuts:

$$\left. \frac{d\sigma_{\text{Born}}}{d\phi} \right|_{\text{data}} = \frac{1}{RC(\phi)} \left. \frac{d\sigma_{\text{exp}}}{d\phi} \right|_{\text{data}}. \quad (6)$$

6 Practical remarks

- **Identical generation limits.** σ_{tot} is integrated over the generation region, so both runs must use exactly the same input limits (Q^2 , E' , W ranges); otherwise the ratio $\sigma_{\text{rad}}/\sigma_{\text{norad}}$ is meaningless.
- **Reconstruct kinematics as in the data.** For radiated events, ϕ , Q^2 , W must be computed from the scattered electron (and detected hadrons), not from the true hadronic-vertex values. The difference between the two is precisely the radiative migration being corrected.
- **The exclusivity cut defines the correction.** The size of RC depends on how much of the radiative tail the missing-mass (or W) cut accepts. The cut applied to the Monte Carlo must match the cut applied to the data exactly; see Sec. 4 for the internal `mm_cut` parameter of `aao_rad`.
- **Generator level vs. after detector simulation.** Taking the ratio after GEMC is legitimate: the acceptance largely cancels in the ratio, and the surviving radiation-induced acceptance change is genuinely part of the correction. The cleaner standard split is to take RC from the generator-level ratio with data-equivalent cuts and the acceptance from the radiated sample alone. Either way, do not correct for the acceptance twice.
- **Bin in the other variables.** RC depends on Q^2 and W , so an inclusive $RC(\phi)$ is an acceptance-weighted average; evaluate it within the (Q^2, W) or (Q^2, t) bins of the analysis.
- **Same polarization flag.** `aao_rad` modulates the cross section with polarization and interference terms; use the same `flag_ehel` in both runs.

- **Statistics.** With 100 000 events and 20 ϕ bins one has ~ 5000 events per bin at generator level, i.e. $\sim 2\%$ statistical uncertainty on RC per bin. After detector acceptance the available statistics may be substantially smaller; generate more events if the ratio is formed after GEMC.

7 Summary

$$\boxed{RC(\phi) = \frac{\sigma_{\text{rad}}}{\sigma_{\text{norad}}} \cdot \frac{h_{\text{rad}}(\phi)}{h_{\text{norad}}(\phi)} \quad \Longrightarrow \quad \sigma_{\text{Born}} = \frac{\sigma_{\text{exp}}}{RC}} \quad (7)$$

where the histograms are filled with equal numbers of unweighted events under identical generation limits and identical analysis cuts.